

A Dynamic General Equilibrium Model of Roads

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1.1 Representative Household

The economy is populated by a representative, infinitely-lived household that maximizes utility over consumption¹, housing services, and local amenities while disliking labor (including commuting time).

Preferences:

$$U = \sum_{t=0}^{\infty} \beta^t \tilde{u}(c_t, h_t, n_t, G_t), \quad 0 < \beta < 1, \quad (1)$$

where c_t denotes non-housing consumption, h_t the flow of housing services, and $A(G_t)$ the amenity derived from the public capital stock G_t , which represents the quality of local public goods.

We consider the following utility function,

$$\tilde{u}(c_t, h_t, n_t, G_t) = c_t h_t^{\alpha_h A(G_t)} \exp\left[-\frac{\kappa}{1+\theta} (n_t [1 + \phi(G_t)])^{1+\theta}\right] \quad (2)$$

After taking logs, we can write the utility function as a convenient separable specification

$$u(c_t, h_t, n_t, G_t) = \ln c_t + \alpha_h A(G_t) \ln h_t - \frac{\kappa}{1+\theta} [n_t (1 + \phi(G_t))]^{1+\theta}, \quad (3)$$

with n_t hours of market work, $\phi(G_t) \geq 0$ a commuting factor that rises as road quality falls, $\kappa > 0$, and $\theta > 0$ is the inverse Frisch elasticity of labor supply i.e. $\frac{1}{\theta}$ is the Frisch elasticity of labor supply with respect to real wages. The parameter α_h captures the household's preference for housing services, and $A(G_t)$ is a function that captures how public infrastructure quality (e.g., roads) affects the utility derived from housing. The term $\phi(G_t)$ captures the disutility of commuting, which increases as road quality deteriorates.

Budget Constraint and Capital Accumulation:

Let w_t be the real wage, r_t the rental rate on private capital K_t , $p_{h,t}$ the price of housing services, τ_L the labor-income tax rate, and τ_H the (voted) property-tax rate. The household's per-period budget constraint is

$$c_t + p_{h,t} h_t + i_t = (1 - \tau_L) w_t n_t + r_t K_t - \tau_H p_{h,t} h_t, \quad (4)$$

¹Since we don't have any random variables or distributions, we are not dealing with any expectations or expected utility.

where i_t denotes investment, and the private-capital law of motion is

$$K_{t+1} = (1 - \delta_K)K_t + i_t, \quad \delta_K \in (0, 1). \quad (5)$$

Optimality Conditions: Maximization of (3) subject to (4) yields:

1. **Consumption–Saving Euler Equation:** The household equalizes the marginal utility of consuming an additional unit today to the discounted utility of saving it until tomorrow. In absence of capital income tax, this yields:

$$U_c(c_t, h_t, A(G_t)) = \beta U_c(c_{t+1}, h_{t+1}, A(G_{t+1})) \left[(1 - \tau_L)w_{t+1} \frac{\partial n_{t+1}}{\partial K_{t+1}} + r_{t+1} + 1 - \delta_K \right]. \quad (6)$$

which simplifies (under certainty and taking labor choices as separate) to

$$U_c(c_t, \cdot) = \beta U_c(c_{t+1}, \cdot) (1 + r_{t+1} - \delta_K) \quad (7)$$

in steady state. In other words, the marginal rate of intertemporal substitution equals the $(1 + \text{net return on capital})$ factor. This is the standard Euler equation ensuring optimal capital accumulation.

Moreover,

$$\frac{1}{c_t} = \beta \frac{1}{c_{t+1}} (1 + r_{t+1} - \delta_K); \quad (8)$$

2. **Labor–leisure trade–off:** The household chooses labor n_t such that the marginal disutility of working (including commuting) equals the after-tax marginal benefit of earning wages. Formally,

$$-U_n(c_t, h_t, A(G_t)) = (1 - \tau_L)w_t U_c(c_t, h_t, A(G_t)) \quad (9)$$

where U_n is the partial derivative of utility with respect to labor (negative due to disutility). Using our utility form,

$$U_n = -\kappa [n_t(1 + \phi(G_t))]^\theta (1 + \phi(G_t)), \quad U_c = \frac{1}{c_t}. \quad (10)$$

Thus, the condition becomes:

$$\kappa [n_t(1 + \phi(G_t))]^\theta (1 + \phi(G_t)) = \frac{(1 - \tau_L)w_t}{c_t}. \quad (11)$$

$$\kappa (1 + \phi(G_t))^{1+\theta} n_t^\theta = \frac{(1 - \tau_L)w_t}{c_t}; \quad (12)$$

$$n_t = \left[\frac{(1 - \tau_L) w_t}{\kappa c_t (1 + \phi(G_t))^{1+\theta}} \right]^{1/\theta} \quad (13)$$

Intuitively, the marginal rate of substitution between leisure (or time) and consumption equals the real after-tax wage. An increase in commuting factor $\phi(G_t)$ (worse roads) raises the left-hand side (effective disutility of labor), leading the household to supply less labor for a given wage. This aligns with the idea that poor road quality reduces effective labor supply (households require higher compensation to work the same hours because commuting erodes their usable time).

3. Housing demand

The household chooses housing h_t such that the marginal utility of housing relative to consumption equals its relative price. The FOC for h_t is:

$$U_h(c_t, h_t, A(G_t)) = U_c(c_t, h_t, A(G_t))(p_{h,t} + \tau_H p_{h,t}), \quad (14)$$

or equivalently,

$$U_h = U_c \cdot (1 + \tau_H) p_{h,t}. \quad (15)$$

Here $p_{h,t}$ is the price of housing and $\tau_H p_{h,t}$ is the property tax per unit housing. In equilibrium, we can define the after-tax housing price as $q_t \equiv (1 + \tau_H) p_{h,t}$. Then the condition becomes:

$$U_h = U_c \cdot q_t. \quad (16)$$

For example, with $U_h = \frac{\alpha_h A(G_t)}{h_t}$ and $U_c = \frac{1}{c_t}$, this condition gives:

$$\frac{\alpha_h A(G_t)}{h_t} = \frac{1}{c_t} q_t, \quad (17)$$

or equivalently:

$$\alpha_h A(G_t) c_t = q_t h_t. \quad (18)$$

This means the household equates the marginal utility per dollar spent on housing to that of consumption. Intuitively, households will demand housing until the point where the utility gain from an extra unit of housing (which diminishes as h_t increases due to diminishing returns) just equals the utility gain from spending those resources on general consumption.

$$\frac{\alpha_h A(G_t)}{h_t} = \frac{(1 + \tau_H)p_{h,t}}{c_t} \implies p_{h,t} = \frac{\alpha_h A(G_t)c_t}{(1 + \tau_H)h_t}. \quad (19)$$

1.2 Representative Firm

A competitive firm produces output Y_t with Cobb–Douglas technology

$$Y_t = K_t^{\alpha_k} (A_t L_t)^{1-\alpha_k}, \quad 0 < \alpha_k < 1, \quad (20)$$

enforcing factor–price equalization

$$w_t = (1 - \alpha_k) \frac{Y_t}{L_t}, \quad r_t = \alpha_k \frac{Y_t}{K_t}. \quad (21)$$

Profit Maximization. The firm operates in a competitive market for inputs and output, so it hires labor and rents capital until factor prices equal their marginal products. In each period, the firm solves

$$\max_{K_t, L_t} \{F(K_t, L_t) - w_t L_t - r_t K_t\}, \quad (22)$$

taking wage w_t and capital rental r_t as given. The first-order conditions are:

1. Labor demand:

$$w_t = (1 - \alpha_k) K_t^{\alpha_k} (A_t L_t)^{-\alpha_k} = (1 - \alpha_k) \frac{Y_t}{L_t}. \quad (23)$$

This is the usual result that the real wage equals the marginal product of labor (MP_L).

2. Capital demand:

$$r_t = \alpha_k K_t^{\alpha_k - 1} (A_t L_t)^{1-\alpha_k} = \alpha_k \frac{Y_t}{K_t}. \quad (24)$$

Thus, the rental rate on capital equals the marginal product of capital (MP_K).

Under constant returns and competitive markets, the firm earns zero economic profit in equilibrium (all output is paid out to factors). This means

$$Y_t = w_t L_t + r_t K_t \quad (25)$$

in equilibrium. Because poor local roads can indirectly reduce effective labor supplied (households may choose smaller L_t) or even the productivity of labor (if A_t were to depend on G_t), the equilibrium wage and output will adjust accordingly. In our model, we focus on the household-side mechanism; we do not explicitly insert G_t into $F(\cdot)$, but one could imagine an extension where $A_t = A(G_t)$, making public infrastructure a productive externality that boosts TFP. That would reinforce the mechanism: a decline in G_t would lower productivity and thus lower wages from the firm side as well. For now, wages change endogenously mainly due to labor supply shifts.

1.3 Government and Public Infrastructure

Taxation and Spending. The government collects taxes and uses the revenue to finance road maintenance. Taxes have two components: an exogenous property tax τ_H (subject to voter approval) and an endogenous tax (such as τ_L on labor income, or potentially a lump-sum tax) that provides additional revenue. Denote by T_t^{exo} the exogenous tax revenue from the voted levy, and T_t^{endo} the endogenous tax revenue from other sources. For example, if τ_L is a labor tax, then $T_t^{\text{endo}} = \tau_L w_t n_t$, which varies with the economy and is therefore “endogenous”. The property tax revenue is $T_t^{\text{exo}} = \tau_H p_{h,t} h_t$ each period, directly tied to the housing market outcome.

The government budget constraint is:

$$T_t^{\text{endo}} + T_t^{\text{exo}} = M_t, \quad (26)$$

assuming all tax revenue is spent on road maintenance M_t . We abstract from any other government consumption or transfers.

In a scenario where the road tax is cut (e.g., τ_H drops to zero after a failed levy vote), T_t^{exo} falls exogenously. For a balanced budget, maintenance spending M_t must be reduced by an equal amount. This captures the immediate fiscal impact of the tax cut, such as an observed 11% loss in road maintenance funding empirically.

Public Capital (Roads) Accumulation. The stock of public road infrastructure G_t evolves over time depending on maintenance. We adopt a law of motion in line with Rioja (2003) to formalize how insufficient maintenance leads to infrastructure depreciation, whereas sufficient maintenance can maintain road quality². Let δ_G be the “normal” depreciation rate of roads (wear-and-tear if adequately maintained). Actual effective depreciation $\delta_{G,t}$ can exceed δ_G if maintenance is below the requirement. A convenient formulation is:

²In this setup, roads can only be maintained, not improved.

$$G_{t+1} = G_t(1 - \delta_{G,t}), \quad (27)$$

where

$$\delta_{G,t} = \varphi \cdot \max\left\{0, 1 - \frac{M_t}{\delta_G G_t^\psi}\right\}. \quad (28)$$

In words, if maintenance spending M_t falls short of the amount needed to cover normal depreciation ($\delta_G G_t$), the depreciation rate increases proportionally to the shortfall. The parameter $\varphi \geq 0$ governs how sensitive the infrastructure is to under-maintenance. For example, if M_t is only half of $\delta_G G_t$, then $1 - \frac{M_t}{\delta_G G_t} = 0.5$, and $\delta_{G,t} = 0.5\varphi$; this means roads wear out. If M_t is zero (no upkeep), depreciation could jump substantially (if φ is large, G_t may deteriorate very quickly). On the other hand, if M_t is at least $\delta_G G_t$ (maintenance fully covers yearly wear), then $\delta_{G,t} = 0$ (no deterioration). We also assume $\delta_{G,t}$ cannot go below 0 even if M_t exceeds $\delta_G G_t$ (any extra maintenance beyond replacing depreciation might marginally improve roads but with diminishing returns). This Riojas-style law of motion captures the intuition that lack of maintenance shortens the life of existing public capital. Maintenance spending effectively slows down depreciation, preserving the infrastructure stock; cutting maintenance causes G_t to decline faster over time. Importantly, a reduction in the road tax τ_H translates to lower M_t and thus a gradual decline in G_t over several periods. The deterioration in road quality will typically become noticeable after a few periods of under-maintenance, consistent with evidence that the negative effects on road conditions (and subsequently house prices) emerge with a lag. In our model, that lag is endogenously determined by the above law of motion – if M_t drops at $t = 0$, G_t will depreciate a bit faster each period, compounding into a significant drop in infrastructure quality after some years.

1.4 Housing Market

Housing is treated as a tradable asset/consumption good in fixed supply. Let $\bar{H}$ denote the total housing stock (or housing services) available in the economy. We assume $\bar{H}$ is exogenously fixed (no new construction, to focus on valuation of a given stock). The Walrasian housing market clears each period, meaning the representative household's demand for housing equals the available supply:

$$h_t = \bar{H}, \quad (29)$$

for all t . Because we have a representative household (or equivalently identical households each demanding h_t and a total supply $\bar{H}$ per household), market clearing effectively pins down h_t in equilibrium. The house price $p_{h,t}$ adjusts to ensure this equality. In other words,

$p_{h,t}$ is determined such that the household's first-order condition for housing (derived above) holds when $h_t = \bar{H}$. Using $U_h/U_c = (1 + \tau_H)p_{h,t}$, we find:

$$(1 + \tau_H)p_{h,t} = \frac{U_h(c_t, h_t, A(G_t))}{U_c(c_t, h_t, A(G_t))} \Big|_{h_t=\bar{H}}, \quad (30)$$

which implicitly gives the equilibrium price $p_{h,t}$ as a function of $\bar{H}$ and other endogenous variables. Intuitively, $p_{h,t}$ will be high if housing is very valuable at the margin (e.g., large α_h or high $A(G_t)$ making housing attractive), and will fall if housing becomes less desirable (for example, if $A(G_t)$ drops due to road decay, lowering U_h relative to U_c). Thus, a decline in road quality G_t reduces the amenity value $A(G_t)$ and increases commuting disutility, both of which make housing less attractive to the household, thereby lowering the equilibrium house price $p_{h,t}$. This is the channel through which the public infrastructure shock is capitalized into property values, as described by Oates's intuition on local public goods and house prices.

1.5 Competitive Equilibrium

We now define a recursive competitive equilibrium for this economy. Given an initial private capital stock K_0 and initial public capital (road quality) G_0 , and a sequence of exogenous tax policy $\{\tau_{H,t}\}$ (with a drop in τ_H at the time of the tax cut shock), a competitive equilibrium is a sequence of allocations $\{c_t, h_t, n_t, K_{t+1}, L_t, M_t, G_t\}_{t \geq 0}$ and prices $\{w_t, r_t, p_{h,t}\}_{t \geq 0}$ such that:

1. **Household Optimization:** Given prices and taxes, the representative household chooses $\{c_t, h_t, n_t, K_{t+1}\}$ to maximize its utility subject to its budget and accumulation constraints. The household's choice satisfies the first-order optimality conditions described above (Euler equation, labor supply condition, and housing demand condition, etc.), and the transversality condition on capital holding.
2. **Firm Optimization:** The representative firm chooses inputs $\{K_t, L_t\}$ each period to maximize profit. In equilibrium, $L_t = n_t$ (labor market clears) and $K_t = K_t$ held by the household (capital market clears), and the wage and rental rates satisfy $w_t = F_L(K_t, L_t)$ and $r_t = F_K(K_t, L_t)$ as given by the Cobb-Douglas marginal products. This ensures the firm's first-order conditions are met and profit is zero.
3. **Government Budget and Public Capital:** The government budget constraint holds each period: $T_t^{\text{endo}} + T_t^{\text{exo}} = M_t$, with $T_t^{\text{exo}} = \tau_{H,t}p_{h,t}h_t$ and $T_t^{\text{endo}} = \tau_L w_t n_t$. The road maintenance M_t feeds into the law of motion for G_t : given G_t at the start of period t , the next period's public capital G_{t+1} is determined by $G_{t+1} = G_t(1 - \delta_{G,t})$ with $\delta_{G,t}$ increasing if M_t is insufficient (as specified above). The government sets M_t according to the available tax revenue each period.

4. **Market Clearing:** All markets clear in equilibrium:

- **Goods market:** Output is used for private consumption, private investment, and public maintenance. The resource constraint each period is:

$$Y_t = c_t + i_t + M_t. \quad (31)$$

Total output Y_t produced by the firm equals the sum of household consumption c_t , private investment i_t , and government demand for road maintenance M_t .

- **Labor market:** $L_t = n_t$. The labor supplied by the household equals labor demanded by the firm. The real wage adjusts such that this holds.
- **Capital market:** The firm's capital input equals the household's capital stock: K_t (used in production) = K_t (owned by household). The rental rate r_t adjusts to clear this market.
- **Housing market:** $h_t = \bar{H}$. The total housing demanded by the household equals the fixed housing stock. The house price $p_{h,t}$ adjusts each period to clear the housing market.

5. **Consistency:** The expectations of the household are consistent with realized outcomes. In a deterministic steady-state scenario, this means the household correctly anticipates the path of prices and G_t . In a perfect-foresight or rational expectations equilibrium, the household foresees that a cut in τ_H implies lower future M_t and thus a gradually declining G_t , factoring this into its decisions. The sequence $\{p_{h,t}\}$ reflects the present value of housing services given the expected decline in amenities.

1.6 Solving the model

The steady-state equilibrium can be characterized as a system of 9 equations in 9 unknowns. We solve for the following endogenous variables: $(c, n, K, Y, w, r, q, M, G)$, where $q \equiv (1 + \tau_H)p_h$ is the effective housing price and housing quantity is fixed at $h = \bar{H}$.

1.6.1 System of Equilibrium Equations

The steady-state equilibrium is determined by the following 9 independent equations:

1. **Euler equation (steady-state):** From the intertemporal optimality condition:

$$1 = \beta(1 + r - \delta_K) \quad \Rightarrow \quad r = \delta_K + \beta^{-1} - 1 \quad (32)$$

This pins down the rental rate r directly from parameters.

2. **Firm capital pricing:** From profit maximization:

$$r = \alpha_k \frac{Y}{K} \quad (33)$$

This relates the output-capital ratio to the rental rate.

3. **Firm wage pricing:** With labor market clearing $L = n$:

$$w = (1 - \alpha_k) \frac{Y}{n} \quad (34)$$

This relates the wage to output and labor supply.

4. **Production technology:** The Cobb-Douglas production function:

$$Y = K^{\alpha_k} (A \cdot n)^{1-\alpha_k} \quad (35)$$

This ties together output, capital, and labor.

5. **Labor-leisure trade-off:** From household optimization:

$$\kappa[1 + \phi(G)]^{1+\theta} n^\theta = \frac{(1 - \tau_L)w}{c} \quad (36)$$

This links labor supply to wages, consumption, and road quality through commuting costs.

6. **Housing demand:** Using housing market clearing $h = \bar{H}$:

$$q = \frac{\alpha_h A(G)c}{\bar{H}} \quad (37)$$

This determines the housing price from consumption and amenities.

7. **Government budget constraint:** Balanced budget condition:

$$M = \tau_L w n + \tau_H \frac{q}{1 + \tau_H} \bar{H} \quad (38)$$

This defines maintenance spending from tax revenues.

8. **Road quality steady-state:** From the law of motion $G_{t+1} = G_t(1 - \delta_{G,t})$:

$$G = G(1 - \delta_G) \quad \text{where} \quad \delta_G = \max \left\{ 0, \varphi \left(1 - \frac{M}{\delta_G^{\text{norm}} G^\psi} \right) \right\} \quad (39)$$

In the minimal-upkeep steady state, this simplifies to $M = \delta_G^{\text{norm}} G^\psi$.

9. **Resource constraint:** Goods market clearing with capital accumulation:

$$c = Y - \delta_K K - M \quad (40)$$

This accounts for the allocation of output between consumption, investment, and maintenance.

1.6.2 Solution Algorithm

The system can be solved using the following iterative approach:

1. **Initialize:** Start with an initial guess for the capital stock $K^{(0)}$ and road quality $G^{(0)}$.
2. **Solve for prices:** Given $K^{(0)}$, use equation (1) to obtain r . This is parameter-determined and independent of other variables.
3. **Inner loop for labor and output:** For given (K, G) , solve the subsystem:
 - Use equations (2) and (4) to express Y in terms of K and n : $Y = K^{\alpha_k}(A \cdot n)^{1-\alpha_k}$
 - From equation (2): $K = \frac{\alpha_k Y}{r}$, which gives us the capital-output relationship
 - Substitute into the production function to solve for n and Y simultaneously
 - Use equation (3) to get $w = (1 - \alpha_k)Y/n$
4. **Solve for consumption and housing price:**
 - Use equation (9) to get $c = Y - \delta_K K - M$
 - But M depends on c through the housing price, so we need to solve equations (5), (6), and (7) simultaneously
 - From equation (5): $c = \frac{(1-\tau_L)w}{\kappa[1+\phi(G)]^{1+\theta}n^\theta}$
 - From equation (6): $q = \frac{\alpha_h A(G)c}{\bar{H}}$
 - From equation (7): $M = \tau_L w n + \frac{\tau_H q \bar{H}}{1+\tau_H}$

5. **Update road quality:** Use equation (8) with the computed M to find the steady-state G that satisfies the road maintenance condition.
6. **Check convergence:** Compare the computed values with the initial guesses. If $|K^{(1)} - K^{(0)}| + |G^{(1)} - G^{(0)}| < \epsilon$ for some small tolerance ϵ , stop. Otherwise, update $(K^{(0)}, G^{(0)}) = (K^{(1)}, G^{(1)})$ and return to step 2.
7. **Verify equilibrium:** Ensure all market clearing conditions and optimality conditions are satisfied with the computed solution.

This algorithm exploits the recursive structure where some variables (like r) are determined purely by parameters, while others require simultaneous solution of subsystems. The convergence is typically rapid due to the well-behaved nature of the log-utility and Cobb-Douglas specifications.

For the transition dynamics following a tax cut, we solve this steady-state system before and after the policy change, then use a perfect foresight shooting algorithm to trace the dynamic path between the two equilibria.

1.7 Transitional Dynamics and Welfare Analysis

To analyze the transitional dynamics following a tax cut, we solve the model numerically under perfect foresight. The key steps are:

1. **Calibration:** Assign parameter values based on empirical estimates or literature benchmarks. For example, β , α_h , η , κ , θ , δ_K , δ_G , and φ are calibrated to match observed household behavior, infrastructure depreciation rates, and labor supply elasticities.
2. **Initial Steady State:** Compute the pre-tax-cut steady state by solving the equilibrium conditions under the initial τ_H . This provides baseline values for $\{c_t, h_t, n_t, K_t, G_t, p_{h,t}, w_t, r_t\}$.
3. **Shock Implementation:** Introduce the tax cut by reducing τ_H at $t = 0$. Solve the model forward to trace the path of endogenous variables $\{G_t, p_{h,t}, c_t, n_t, K_t\}$ as the economy transitions to the new steady state.
4. **Welfare Analysis:** Compute the household's lifetime utility before and after the tax cut. Decompose the welfare change into short-run gains (higher disposable income) and long-run losses (lower amenities and higher commuting costs).

5. **Sensitivity Analysis:** Vary key parameters (e.g., φ , α_h , θ) to test the robustness of the results. Examine how the magnitude of the tax cut or alternative government policies (e.g., lump-sum taxes) affect outcomes.

The numerical solution involves iterating on the household's Euler equations, firm conditions, and government budget constraint while updating G_t using its law of motion. This allows us to simulate the gradual decline in road quality and its impact on housing prices, labor supply, and overall welfare.

1.8 Discussion of Policy Shock

A permanent cut in the voted levy lowers τ_H exogenously. In the short run, disposable income rises as tax burden decreases, but G_t is initially unchanged. This leads to a sharp increase in consumption but also a sharp decrease in maintenance budget. Over time, reduced maintenance spending accelerates depreciation in equation 27, lowering $A(G_t)$ and raising $\phi(G_t)$; this depresses labor supply, housing demand, and therefore equilibrium house prices, as demonstrated in Section 1.10.

1.9 Model Calibration

The model parameters are calibrated based on empirical evidence and standard values from the literature. Table 1 summarizes the calibrated values used in the analysis. These values are chosen to reflect realistic economic conditions. For example, $\beta = 0.96$ corresponds to a 4% annual discount rate, while $\delta_K = 0.05$ reflects typical depreciation rates for private capital. The labor disutility parameters κ and θ are calibrated to match observed labor supply elasticities, and φ captures the empirical sensitivity of road quality to maintenance shortfalls.

1.10 Model Results

The model is solved numerically using a perfect foresight algorithm to simulate the economy's response to a permanent cut in the road maintenance tax τ_H . The key results are summarized below:

- A reduction in τ_H from 2.5% to 0% leads to an decrease in road quality G_t , which declines gradually over time due to the law of motion for public capital.
- The house price $p_{h,t}$ falls significantly as the amenity value $A(G_t)$ declines, reflecting the reduced desirability of housing in areas with deteriorating roads

Table 1: Baseline calibration and sensitivity ranges

Parameter	Model block	Baseline	Sensitivity range	Key empirical source
β	Household intertemporal utility	0.96	0.94–0.99	Cooley (1995)
α_h	utility weight on housing services	0.35	0.25–0.40	(U.S. Bureau of Labor Statistics, 2024)
η	Amenity elasticity	0.42		†
θ	Inverse Frisch elasticity labor supply	2	1.5–4.0	Chetty et al. (2011)
κ	Labor-disutility scale	2.9	2.9	Chatterjee, Gibson and Rioja (2017)
α_k	Private-capital share	0.30	0.25–0.40	Chatterjee, Gibson and Rioja (2017)
δ_K	Private Capital depreciation (annual)	0.060	0.04–0.08	Cooley (1995)
δ_G^{norm}	Road stock depreciation	0.04	0.015–0.060	(Kornfeld and Fraumeni, 2022)
φ	Under-maintenance sensitivity	0.64	0.25–0.75	Rioja (2003)
γ	commuting disutility sensitivity	0.057		†
ψ	maintenance requirement elasticity	0.45		†
τ_L	Labor income tax share for roads	0.046		†
<i>Note:</i> † Calibrated to match empirical moments				

The figures 1 illustrates the dynamics of key variables over time following the tax cut. The model's transitional dynamics following a permanent reduction in the property tax from 1.5 mills to 0 mills.

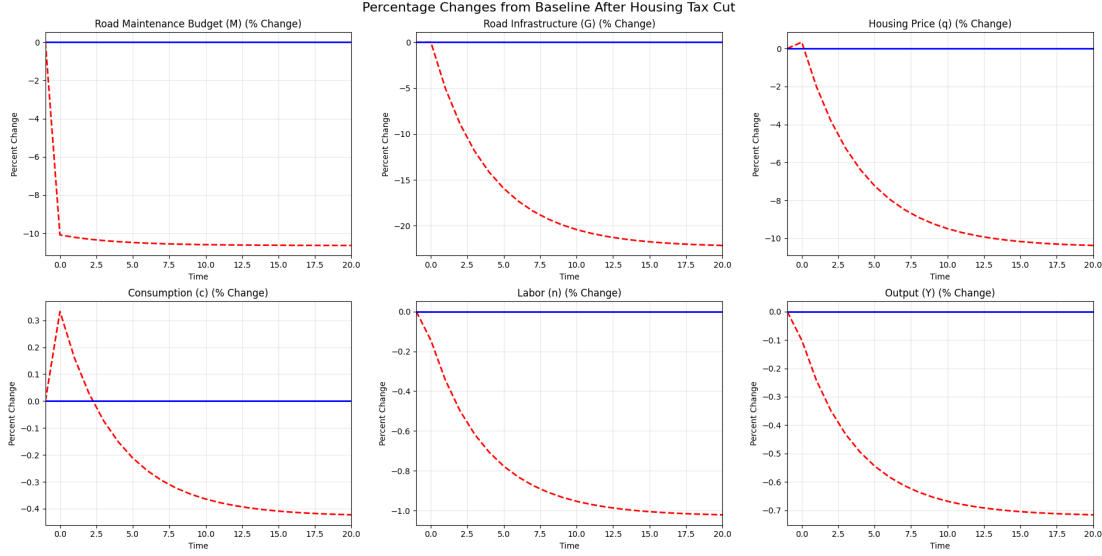


Figure 1: Dynamics of variables following the tax cut.

Figure 1 plots the full transition paths (percentage deviations from baseline). Most of the adjustment in M_t and q_t occurs early, while G_t converges more gradually as under-maintenance compounds over time. Findings from Figure 1 are summarized below:

- **Roads and maintenance.** Following the road tax cut, the road maintenance budget drops by about 10% and then drifts to roughly -11% (top-left panel of Figure 1). Given the law of motion for roads, G_t declines gradually and monotonically, approaching a long-run decline to 23% by the end of the simulated horizon.
- **House prices.** The housing price q_t initially shows a small blip (about $+0.3\%$) and then declines smoothly toward a long-run drop of an average of 9% over the simulated horizon as the amenity value $A(G_t)$ deteriorates.
- **Aggregates.** The macro aggregates move modestly: labor n_t falls to about -1.0% , output Y_t to about -0.7% , and consumption c_t to about -0.4% in the long run. Consumption exhibits a slight short-run overshoot (about $+0.3\%$) before settling below baseline (bottom row). This reflects the household's initial response to higher disposable income from the tax cut, followed by adjustments as road quality declines.

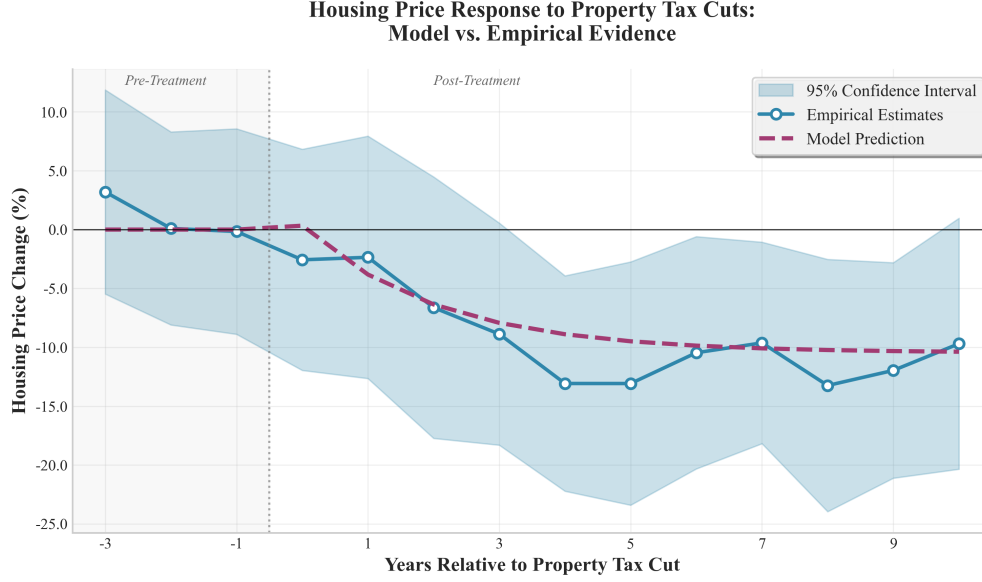


Figure 2: Model vs Empirical Estimates of House Prices Following the Tax Cut.

Model vs. empirical house-price path. Figure 2 compares the model-implied house-price response to the event-study estimates in Figure ?? . After aligning the time units (every two model periods corresponds to one event-study year), the model tracks the medium- and long-run decline well, squarely within the empirical confidence band. The model is slightly conservative around the empirical trough—peaking—yielding a root-mean-squared gap of roughly 2.1 percentage points over $t = 0$ to $t = 10$. This pattern is consistent with the mechanism: eliminating the tax reduces maintenance revenues, which gradually lowers road quality, depresses amenity $A(G_t)$, and in turn pulls down housing valuations.

Overall, the calibrated model reproduces the targeted steady-state effects (about -23% in roads, -9% in house prices, and -11% in maintenance) and generates transitional dynamics for house prices that closely follow the empirical path while remaining slightly less sharp at the short-run trough.

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